Program 10:
Demonstrate Kubernetes Dashboard and CLI for Cluster Monitoring
19/12/2025

Project Directory:

prog10
|- deployment.yaml
|- service.yaml

Steps for Execution:

Step 1: Move to the directory where we work for DevOps lab. Then create a project directory with the name prog10

> mkdir prog10
> cd prog10/

```
1RV24MC036_gayatri_v_k@ubuntu:~$ cd program10
1RV24MC036_gayatri_v_k@ubuntu:~/program10$ kubectl get nodes
NAME       STATUS   ROLES           AGE     VERSION
minikube   Ready    control-plane   5m17s   v1.34.0
```

Step 2: Initialize the minikube with specific number of CPUs and memory, enable the kubernetes dashboard by using the command,

> minikube start –cpus=2 --memory=4096
> minikube dashboard

```
docker --version
minikube version: v1.37.0
commit: 65318f4cfff9c12cc87ec9eb8f4cdd57b25047f3
Client Version: v1.34.3
Kustomize Version: v5.7.1
Docker version 28.5.1, build e180ab8
1RV24MC036_gayatri_v_k@ubuntu:~$ minikube delete
🔥  Deleting "minikube" in docker ...
🔥  Deleting container "minikube" ...
🔥  Removing /home/1RV24MC036_GAYATRI_V_K/.minikube/machines/minikube ...
💀  Removed all traces of the "minikube" cluster.
1RV24MC036_gayatri_v_k@ubuntu:~$ minikube start --driver=docker --cpus=2 --memory=4096
😄  minikube v1.37.0 on Ubuntu 24.04
✨  Using the docker driver based on user configuration
📌  Using Docker driver with root privileges
👍  Starting "minikube" primary control-plane node in "minikube" cluster
🚜  Pulling base image v0.0.48 ...
❗  minikube was unable to download gcr.io/k8s-minikube/kicbase:v0.0.48, but successfully downloaded docker.io/kicbase/stable:v0.0.48@sha256:7171c97a51623558720f8e5878e4f4637da093e2f2e
d589997bedc6c1549b2b1 as a fallback image
🔥  Creating docker container (CPUs=2, Memory=4096MB) ...
❗  Failing to connect to https://registry.k8s.io/ from inside the minikube container
💡  To pull new external images, you may need to configure a proxy: https://minikube.sigs.k8s.io/docs/reference/networking/proxy/
🐳  Preparing Kubernetes v1.34.0 on Docker 28.4.0 ...
🔗  Configuring bridge CNI (Container Networking Interface) ...
🔎  Verifying Kubernetes components...
    ▪ Using image gcr.io/k8s-minikube/storage-provisioner:v5
🌟  Enabled addons: storage-provisioner, default-storageclass
🏄  Done! kubectl is now configured to use "minikube" cluster and "default" namespace by default
1RV24MC036_gayatri_v_k@ubuntu:~$ kubectl get nodes
NAME       STATUS   ROLES           AGE   VERSION
minikube   Ready    control-plane   17s   v1.34.0
1RV24MC036_gayatri_v_k@ubuntu:~$ minikube dashboard
🔌  Enabling dashboard ...
    ▪ Using image docker.io/kubernetesui/dashboard:v2.7.0
    ▪ Using image docker.io/kubernetesui/metrics-scraper:v1.0.8
💡  Some dashboard features require the metrics-server addon. To enable all features please run:

        minikube addons enable metrics-server

🤔  Verifying dashboard health ...
🚀  Launching proxy ...
🤔  Verifying proxy health ...
🎉  Opening http://127.0.0.1:40175/api/v1/namespaces/kubernetes-dashboard/services/http:kubernetes-dashboard:/proxy/ in your default browser...
Gtk-Message: 09:44:25.319: Not loading module "atk-bridge": The functionality is provided by GTK natively. Please try to not load it.
```

Step 3: Now create a two files inside the prog10 folder
1. deployment.yaml 2. service.yaml
The content of the above files is in the below images

```
1RV24MC036_gayatri_v_k@ubuntu:~/program10$ nano deployment.yaml
1RV24MC036_gayatri_v_k@ubuntu:~/program10$ nano service.yaml
```

Step 4: The project folder structure looks like below

```
1RV24MC036_gayatri_v_k@ubuntu:~/program10$ ls
deployment.yaml  service.yaml
```

Step 5: Apply both yaml in the current directory of prog10 folder by using the commands,
> kubectl apply -f deployment.yaml
> kubectl apply -f service.yaml

```
1RV24MC036_gayatri_v_k@ubuntu:~/program10$ kubectl get svc
NAME            TYPE        CLUSTER-IP       EXTERNAL-IP   PORT(S)        AGE
kubernetes      ClusterIP   10.96.0.1        <none>        443/TCP        5m23s
nginx-service   NodePort    10.110.255.171   <none>        80:30526/TCP   3m42s
1RV24MC036_gayatri_v_k@ubuntu:~/program10$ kubectl get pods
NAME                               READY   STATUS    RESTARTS   AGE
nginx-deployment-6f9664446b-44nrr   1/1     Running   0          2m59s
nginx-deployment-6f9664446b-pxbt8   1/1     Running   0          2m59s
1RV24MC036_gayatri_v_k@ubuntu:~/program10$ kubectl logs nginx-deployment-6f9664446b-pxbt8
/docker-entrypoint.sh: /docker-entrypoint.d/ is not empty, will attempt to perform configuration
/docker-entrypoint.sh: Looking for shell scripts in /docker-entrypoint.d/
/docker-entrypoint.sh: Launching /docker-entrypoint.d/10-listen-on-ipv6-by-default.sh
10-listen-on-ipv6-by-default.sh: info: Getting the checksum of /etc/nginx/conf.d/default.conf
10-listen-on-ipv6-by-default.sh: info: Enabled listen on IPv6 in /etc/nginx/conf.d/default.conf
/docker-entrypoint.sh: Sourcing /docker-entrypoint.d/15-local-resolvers.envsh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/20-envsubst-on-templates.sh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/30-tune-worker-processes.sh
/docker-entrypoint.sh: Configuration complete; ready for start up
2025/12/19 04:15:21 [notice] 1#1: using the "epoll" event method
2025/12/19 04:15:21 [notice] 1#1: nginx/1.29.4
2025/12/19 04:15:21 [notice] 1#1: built by gcc 14.2.0 (Debian 14.2.0-19)
2025/12/19 04:15:21 [notice] 1#1: OS: Linux 6.8.0-51-generic
2025/12/19 04:15:21 [notice] 1#1: getrlimit(RLIMIT_NOFILE): 1048576:1048576
2025/12/19 04:15:21 [notice] 1#1: start worker processes
2025/12/19 04:15:21 [notice] 1#1: start worker process 29
2025/12/19 04:15:21 [notice] 1#1: start worker process 30
2025/12/19 04:15:21 [notice] 1#1: start worker process 31
2025/12/19 04:15:21 [notice] 1#1: start worker process 32
1RV24MC036_gayatri_v_k@ubuntu:~/program10$ kubectl describe pod nginx-deployment-6f9664446b-pxbt8
Name:             nginx-deployment-6f9664446b-pxbt8
Namespace:        default
Priority:         0
Service Account:  default
Node:             minikube/192.168.49.2
Start Time:       Fri, 19 Dec 2025 09:43:55 +0530
Labels:           app=nginx
                  pod-template-hash=6f9664446b
Annotations:      <none>
Status:           Running
IP:               10.244.0.7
```

Step 6: Now verify all the resources such pas nodes, pods, svc and deployments by using the commands,
> kubectl get nodes
> kubectl get pods
> kubectl get svc
> kubectl get deployments

Step 7: Check all the Pod logs, describe and view all the details of the deployments by using the command,

> kubectl logs <pod-name>

> kubectl describe pod <pod-name>

```
1RV24MC036_gayatri_v_k@ubuntu:~/program10$ kubectl describe pod nginx-deployment-6f9664446b-pxbt8
Name:             nginx-deployment-6f9664446b-pxbt8
Namespace:        default
Priority:         0
Service Account:  default
Node:             minikube/192.168.49.2
Start Time:       Fri, 19 Dec 2025 09:43:55 +0530
Labels:           app=nginx
                  pod-template-hash=6f9664446b
Annotations:      <none>
Status:           Running
IP:               10.244.0.7
IPs:
  IP:           10.244.0.7
Controlled By:  ReplicaSet/nginx-deployment-6f9664446b
Containers:
  nginx:
    Container ID:   docker://3bef4fa59545fb2430de456f42d39e025f01e00d32f186df05db522fc45050a5
    Image:          nginx:latest
    Image ID:       docker-pullable://nginx@sha256:fb01117203ff38c2f9af91db1a7409459182a37c87cced5cb442d1d8fcc66d19
    Port:           80/TCP
    Host Port:      0/TCP
    State:          Running
      Started:      Fri, 19 Dec 2025 09:45:21 +0530
    Ready:          True
    Restart Count:  0
    Environment:    <none>
    Mounts:
      /var/run/secrets/kubernetes.io/serviceaccount from kube-api-access-7qhl5 (ro)
Conditions:
  Type                        Status
  PodReadyToStartContainers   True
  Initialized                 True
  Ready                       True
  ContainersReady             True
  PodScheduled                True
Volumes:
  kube-api-access-7qhl5:
    Type:                    Projected (a volume that contains injected data from multiple sources)
    TokenExpirationSeconds:  3607
    ConfigMapName:           kube-root-ca.crt
    Optional:                false
```

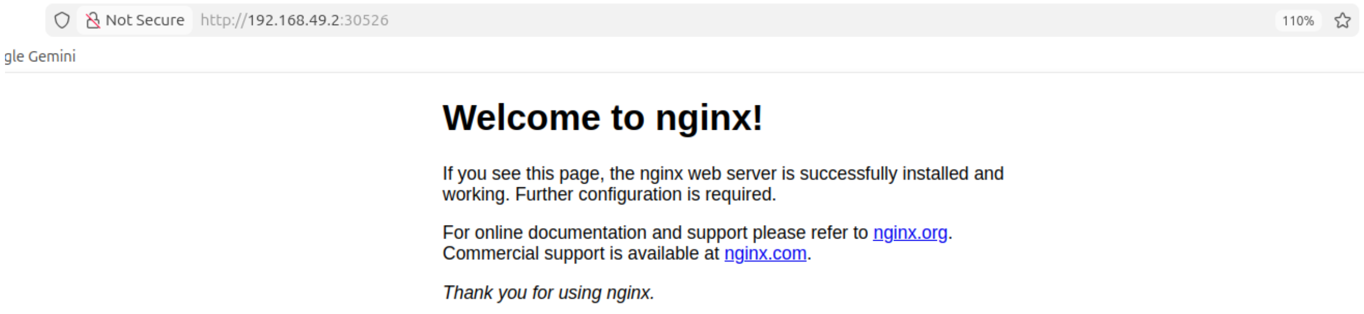
Step 8: We can now access the service in the browser by using the command,
> minikube service nginx-service

```
1RV24MC036_gayatri_v_k@ubuntu:~/program10$ minikube service nginx-service
```

NAMESPACE	NAME	TARGET PORT	URL
default	nginx-service	80	http://192.168.49.2:30526

```
🎉  Opening service default/nginx-service in default browser...
```

Step 9: After performing the above command the browser page will opens automatically with the address, http://192.168.49.2:30008



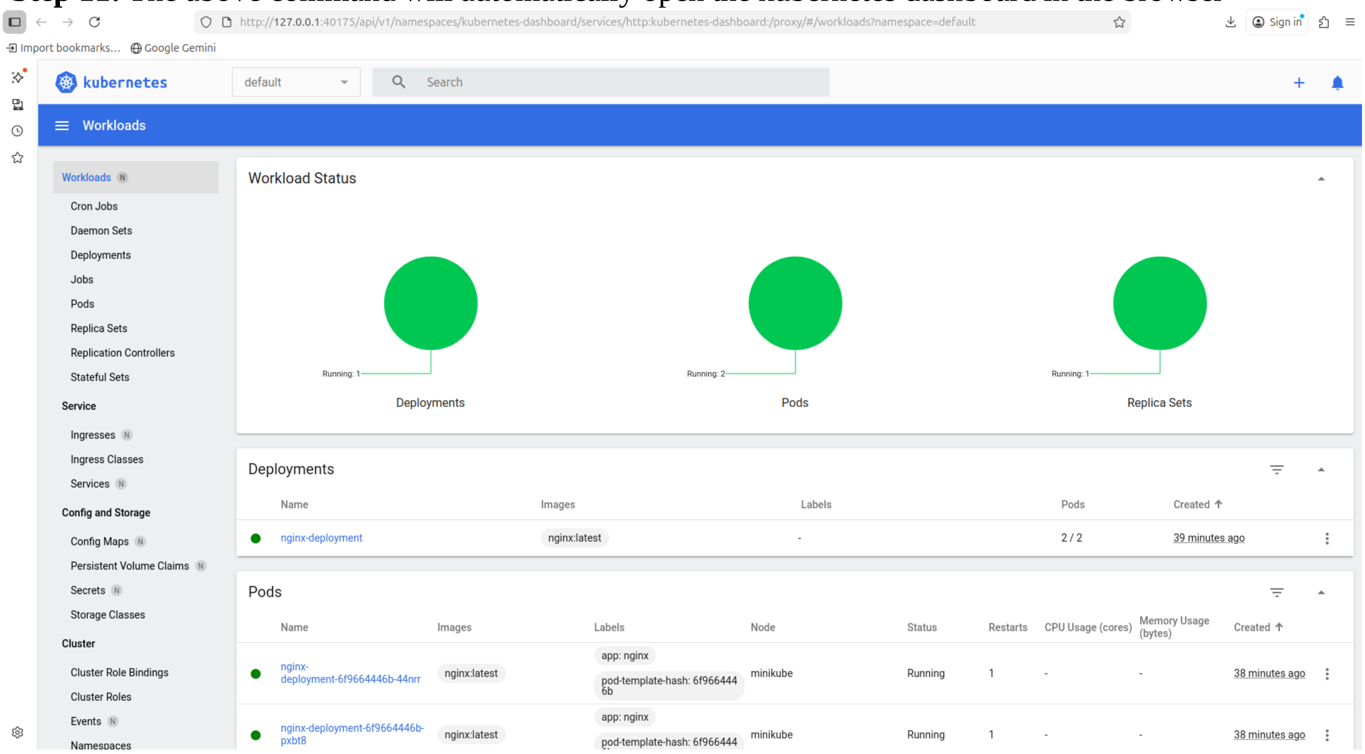
Step 10: Now open the kubernetes dashboard using the command,
> minikube dashboard

```
1RV24MC036_gayatri_v_k@ubuntu:~$ minikube dashboard
🔌  Enabling dashboard ...
    ▪ Using image docker.io/kubernetesui/dashboard:v2.7.0
    ▪ Using image docker.io/kubernetesui/metrics-scraper:v1.0.8
💡  Some dashboard features require the metrics-server addon. To enable all features please run:

        minikube addons enable metrics-server

🤔  Verifying dashboard health ...
🚀  Launching proxy ...
🤔  Verifying proxy health ...
🎉  Opening http://127.0.0.1:40175/api/v1/namespaces/kubernetes-dashboard/services/http:kubernetes-dashboard:/proxy/ in your default browser...
Gtk-Message: 09:44:25.319: Not loading module "atk-bridge": The functionality is provided by GTK natively. Please try to not load it.
```

Step 11: The above command will automatically open the kubernetes dashboard in the browser



Step 12: Now delete or Clean-up all the pods, services and deployments by using commands,
> kubectl delete pods --all
> kubectl delete svc --all
> kubectl delete deployements --all